Boom with Siri
Ultimate Ears adds Google and Siri voice integration to UE Boom 2 and Megaboom Bluetooth speakers. **http://dgit.in/UeSiri**

Gameboy phone?
The Hyperkin SmartBoy allows you to turn your phone into a GameBoy. **http://dgit.in/smartboyx**

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An oft-abused term in the technology arena, to achieve 'wireless nirvana' meant making wired electronic devices stay on and functional without wires. How naive of us that this gradual shift alone would rid us of our gadget trials and tribulations, help us attain nirvana. Pfft! Back then were the early days of the digital age. We were easily swayed by delusions of technical grandeur. We were too hasty in patting our backs by merely eradicating wires than to comprehend the limitations of two things that propelled the wireless age – radio waves and battery – the latter of which continues to be the catalyst of many shattered wireless dreams.

Not anymore, though. Not if WISP can help it.

Well, imagine a microcontroller – a chip – that for all practical purpose powers itself without any batteries or plugged in power source and be receptive to programmable wireless instructions. Sounds something straight out of a futuristic sci-fi movie, isn't it? Not at all. This is the power of WISP, a breakthrough protocol that couldn't have come any sooner to rid us off our wireless woes! The topic of this article – you guessed it – is about looking into what WISP (wireless identification sensing platform) is all about and just why it is that it promises to hold so much potential as far as revolutionizing wireless communications in the coming years.

In the beginning...

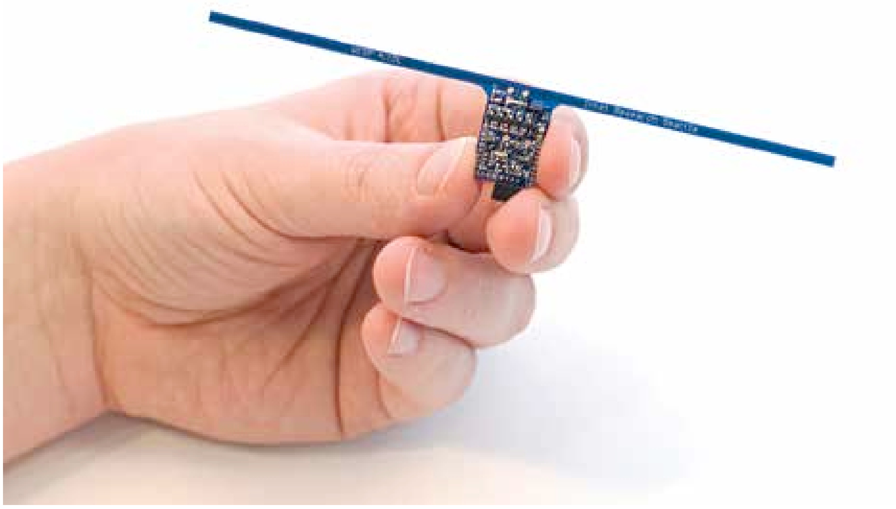
The WISP project grew out of an Intel Labs initiative nearly 12 years ago from the company's office in Seattle. What began as a project to make iterative breakthroughs to steadily progress the cause of wireless communication, J.R. Smith, a lead researcher on the project, describes how it quickly became "the first far-field RF powered sensing and computational platform." Those early days of WISP weren't without successful failures and lots of learnings, according to Smith, who recounts how he inducted more researchers under him from the University of Washington (in Seattle) and even went all the way to fabricating an early chip on which to implement the RFID tag through the Intel Shuttle Program before it was shut down. In 2009, WISP was mentioned under the archive of innovation at Intel.

However, the project wasn't completely on the backburner as far as the University of Washington was concerned, and with the help of TU Delft (the largest and oldest public university of technology) in the Netherlands, they've really unearthed some breakthroughs in WISP.

Wispy magic

So far, RFIDs have always needed two things to work -- power (either in the form of battery or plugged into AC source) and benchmarks, but it's a chip that's several times more powerful and larger than anything researchers have tried to power and program using only radio waves (or ambient energy) so far. That's right, WISP chips are the only ones in the world that can be wirelessly reprogrammed, a feat that other RFID devices that power themselves off stray radio transmissions like television and mobile base stations still cannot do. And for that fact alone, this communication breakthrough is of great significance.

How does it work its magic? Well, the group of researchers from America and the Netherlands who developed the WISP5 sensor attribute the platform's success and its ability to operate at such infinitesimally small packets of energy to the use of a ferroelectric RAM module.



The original Intel prototype developed before the Intel Shuttle Program was shut down

wired connectivity for it to be reprogrammable in any way. And the same holds true for other wide-scale deployments of similar wireless protocol. Both these physical hurdles of battery and wired connectivity are finally being overcome through the latest iteration of WISP, according to new research breakthroughs coming from the joint work of both the American and Dutch university students and faculty.

No longer based fully on the Intel architecture, the RISC-based 16-bit microcontroller that's modestly clocked at 16MHz won't break any performance

How's it crucial? In comparison to NAND modules that make up conventional RAM modules from pretty much every device you can think of – smartphone, PC, gaming console, router, laptop, etc. – a Ferroelectric RAM or FRAM module requires just a tiny fraction of energy, which is about a hundred times lesser. And in the realm of ambient energy powered electronic devices, this is again an important breakthrough.

Where earlier RFID sensors relied upon dedicated batteries or power sources to be in an always ON state or turn ON (based on RF input) to establish a mode of